

assignment conservation energy

 This is a preview of the published version of the quiz

Started: May 7 at 2:38pm

Quiz Instructions

There is no tutorial. But hints instead. . Use a post-it to hide the hints. Try first without looking at the hints.



Question 1 6.25 pts

Two stones, one of mass m and the other of mass $2m$, are thrown directly upward with the same velocity at the same time from ground level and feel no air resistance. Which statement about these stones is true?

hint: *What is the expression for the PE?*



The heavier stone will go twice as high as the lighter one because it initially had twice as much kinetic energy.



At their highest point, both stones will have the same gravitational potential energy because they reach the same height.



Both stones will reach the same height because they initially had the same amount of kinetic energy.



At its highest point, the heavier stone will have twice as much gravitational potential energy as the lighter one because it is twice as heavy.



The lighter stone will reach its maximum height sooner than the heavier one.



Question 2 6.25 pts

A horizontal spring-mass system oscillates on a frictionless table. If the ratio of the mass to the spring constant is 0.023 kgm/N and the maximum speed of the mass was measured to be 24.90 m/s find the maximum extension of the spring.

hint: $PE_{\text{el}} = KE$. Use $0.5 K X^2 = 0.5 m V^2$ and $m/K = 0.023$

☐

3.8 cm

☐

0.57 m

☐

3.8 m

☐

57 cm

☐

Question 3 6.25 pts

If a ball is released from a window ledge, how fast will it be travelling when it reaches the ground, 69m below? (PE = KE. cross out the mass)

☐

37 m/s

☐

304 m/s

☐

74 m/s

☐

1.38 km/s

☐

Question 4 6.25 pts

A block slides down a frictionless inclined ramp. If the ramp angle is 17.0° and the length of it is 20m, find the speed of the block as it reaches the end of the ramp, assuming it started sliding from rest at the top.

hint: $mgh = KE$. Use geometry to find h. In a right angle triangle

☐

7.57 m/s

☐

10.7 m/s

☐

19.6 m/s

☐

114 m/s

☐

Question 5 6.25 pts

Sue and Betti both ski straight down a hill, both starting from rest. Sue weighs more than Betti. Neglecting friction and wind resistance, which skier will be moving the fastest at the bottom of the hill?

☐

They will be moving with the same speed.

☐

Sue

☐

Betti

⋮

Question 6 6.25 pts

Swimmers at a water park have a choice of two frictionless water slides as shown in the figure. Although both slides drop over the same height, h , slide 1 is straight while slide 2 is curved, dropping quickly at first and then leveling out. How does the speed v_1 of a swimmer reaching the end of slide 1 compares with v_2 , the speed of a swimmer reaching the end of slide 2?

hint: gravity is a conservative force. The path from point 1 to 2 does not matter.


☐

$v_1 < v_2$

☐

No simple relationship exists between v_1 and v_2 because we do not know the curvature of slide 2.

☐

$v_1 > v_2$

☐

$v_1 = v_2$

⋮

Question 7 6.25 pts

A box of mass m is pressed against (but is not attached to) an ideal spring of force constant k and negligible mass, compressing the spring a distance x . After it is released, the box slides up a frictionless incline as shown in the figure and eventually stops. If we repeat this experiment but instead use a spring having force constant $2k$

hint: Elastic energy (stage1) = kinetic energy (stage2)=potential energy(stage3).

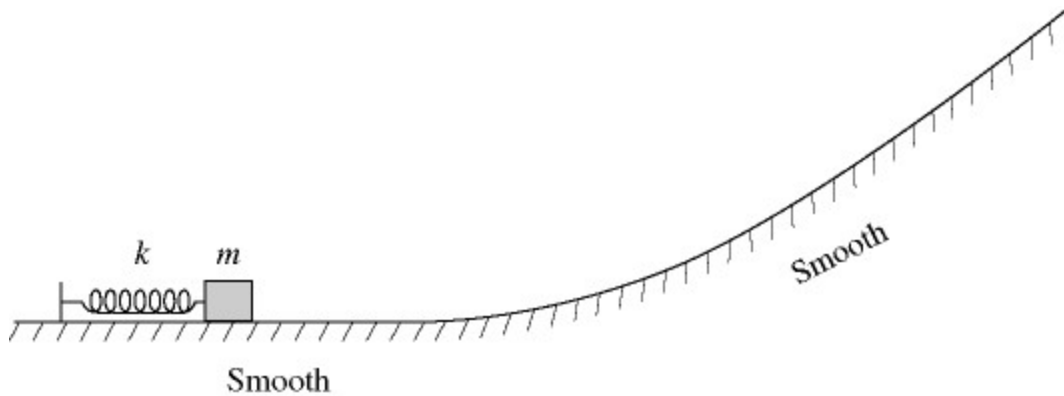
Take a simple example. Say $K=1$, $x=1\text{m}$, $m=1\text{kg}$. Find the speed when it is released and then find the height.

Then take $K=2$, $x=1\text{m}$, $m=1\text{kg}$ What happens to speed ? (is it multiplied by $1.44=\text{square root of } 2$?)

what happens to the kinetic energy? what happens to the height?

so $0.5 K x^2 = 0.5 m V^2$ find V and $0.5 k x^2 = mgh$ find h with $k=1$, $x=1$, $m=1$

Then something with $k=2$ and $x=1$, $m=1$


☐

None of the above choices is correct.

☐

the box will go up the incline twice as high as before.

☐

just as it moves free of the spring, the speed of the box will be $\sqrt{2}$ times as great as before.

☐

just as it moves free of the spring, the kinetic energy of the box will be twice as great as before.

☐

All of the above choices are correct.

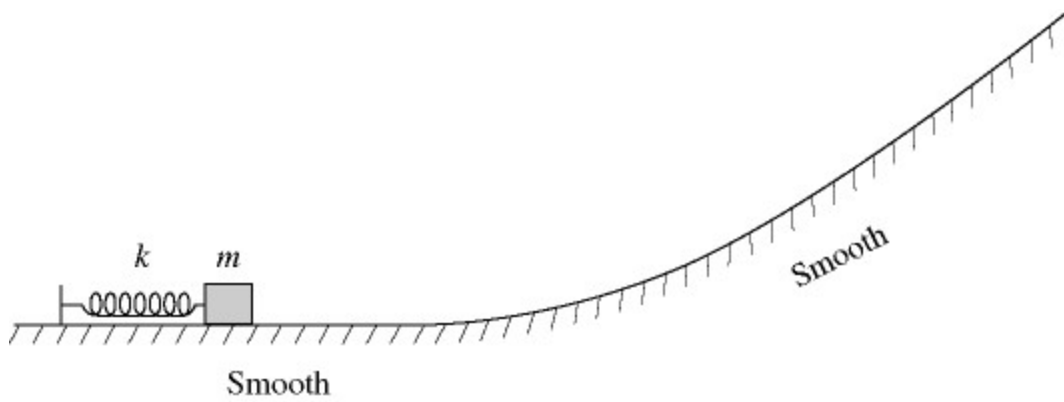
⋮

Question 8 6.25 pts

A box of mass m is pressed against (but is not attached to) an ideal spring of force constant k and negligible mass, compressing the spring a distance x . After it is released, the box slides up a frictionless incline as shown in the figure and eventually stops. If we repeat this experiment but instead compress the spring a distance of $2x$

hint: hint: Elastic energy (stage1) = kinetic energy (stage2)=potential energy(stage3).

Take a simple example. Say $K=1$, $x=1\text{m}$, $m=1\text{kg}$. find the speed after release. find the height. Then see what happens when $K=1$ and $x = 2\text{m}$, $m=1\text{kg}$



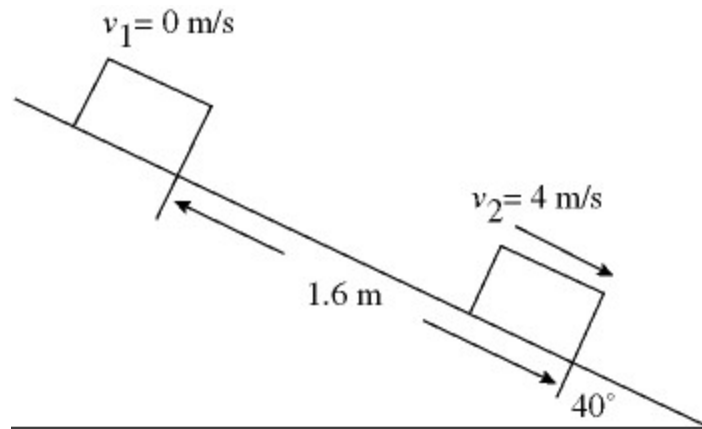
- ☐ just as it moves free of the spring, the box will have twice as much kinetic energy as before.
- ☐ the box will go up the incline twice as high as before.
- ☐ just as it moves free of the spring, the box will be traveling twice as fast as before.
- ☐ just as it moves free of the spring, the box will be traveling four times as fast as before.
- ☐ just before it is released, the box has twice as much elastic potential energy as before.



Question 9 6.25 pts

An 8.0-kg block is released from rest, with $v_1 = 0.00$ m/s, on a rough incline, as shown in the figure. The block moves a distance of 1.6-m down the incline, in a time interval of 0.80 s, and acquires a velocity of $v_2 = 4.0$ m/s. How much work does gravity do on the block during this process?

hint: you have 2 ways to do it. You can use the fact that gravity is a conservative force so the path does not matter. work done by gravity = $mg(\text{different in height})$. To find the height use trig. $\sin(40) = h/1.6\text{m}$ Or you can compute the work done by the component of gravity along the incline which is $mg\sin(40)$. so work is force $\times$ distance. distance is 1.6m . The work is positive. !!! you can not use the change in PE because the surface is rough so the change in PE is not equal to the change in KE. Friction is doing negative work. They only ask for the work done by gravity.



☐ +100 J

☐ +81 J

☐ -100 J

☐ +120 J

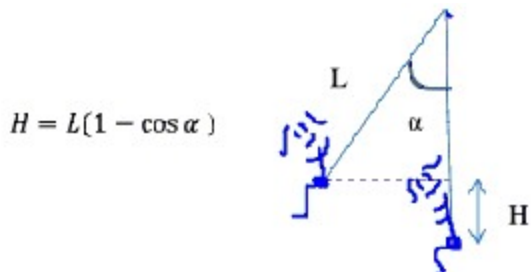
☐ -81 J



Question 10 6.25 pts

You do 174 J of work while pulling your sister back on a swing, whose chain is 5.10 m long. You start with the swing hanging vertically and pull it until the chain makes an angle of 32.0° with the vertical with your sister is at rest. What is your sister's mass, assuming negligible friction?

Make a drawing.



hint: you are doing work against gravity. The work you are doing = - work done by gravity = $mg(H)$.

See above to find H . $H = L - L\cos(32)$

☐ 28.4 kg

☐ 19.5 kg



26.3 kg



22.9 kg



Question 11 6.25 pts

An athlete stretches a spring an extra 40.0 cm beyond its initial length. How much energy has he transferred to the spring, if the spring constant is 52.9 N/cm?

hint: that's easy. convert. 1 kJ = 1000J



423 kJ



4230 kJ



4230 J



423 J



Question 12 6.25 pts

A tennis ball bounces on the floor three times. If each time it loses 22.0% of its energy due to heating, how high does it rise after the third bounce, provided we released it 2.3m from the floor?

hint: initial potential energy is mgh . Each time it bounces up it reaches only 78% its previous height. Since the mass and g don't change. $PE = mgh$. Find the height after each bounce,



140 cm



110 cm



110 mm



11 cm



Question 13 6.25 pts

A 1.86-kg block is held in place against the spring by a 81-N horizontal external force (see the figure). The external force is removed, and the block is projected with a velocity $v_1 = 1.2$ m/s upon separation from the spring. The block descends a ramp and has a velocity $v_2 = 1.9$ m/s at the bottom. The track is frictionless between points *A* and *B*. The block enters a rough section at *B*, extending to *E*. The coefficient of kinetic friction over this section is 0.28. The velocity of the block is $v_3 = 1.4$ m/s at *C*. The

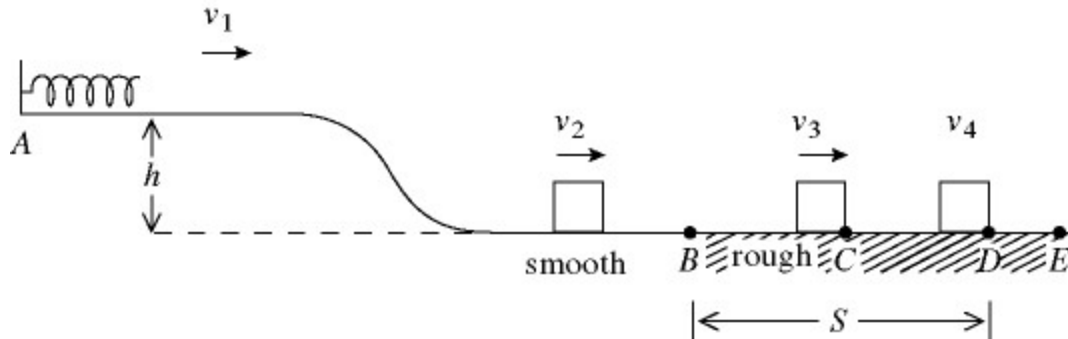
block moves on to D , where it stops. The height h of the ramp is closest to (in cm)

hint: this one is easy. They just make it sound challenging. It goes through stages.

elastic energy + PE = kinetic energy₁ + PE = kinetic energy₂ = kinetic energy₃ + loss

But we just need to use stage 2 and stage 3. that is kinetic energy₁ + potential energy = kinetic energy₂.

So use v_1, h, v_2 . $0.5 m v_1^2 + m g h = 0.5 m v_2^2$ (cross out m)



- ☐ 17
- ☐ 18
- ☐ 11
- ☐ 15
- ☐ 7.3
- ☒

Question 14 6.25 pts

In the figure, a block of mass m is moving along the horizontal frictionless surface with a speed of 5.70 m/s. If the slope is 11.0° and the coefficient of kinetic friction between the block and the incline is 0.260, how far does the block travel up the incline?

hint: conservation of energy. kinetic energy = potential energy + loss due to heat (or kinetic energy - loss = potential energy)

loss = work done by friction = friction $\times$ distance . distance is the unknown.

friction = $\mu_k N$. The normal force is always $mg \cos(\text{angle})$.

potential energy = $mg h$. Use geometry. $h = \sin(\text{angle})$.

So you have one equation with one unknown. cool !



1.



1.25m



2.35m



5m



3.72 m



Question 15 6.25 pts

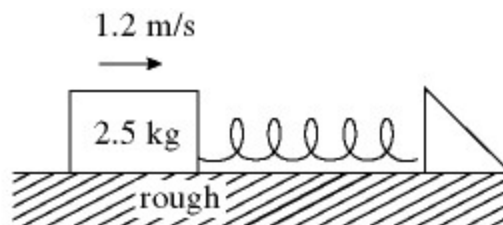
A 2.5-kg box, sliding on a rough horizontal surface, has a speed of 1.2 m/s when it makes contact with a spring (see the figure). The block comes to a momentary halt when the compression of the spring is 5.0 cm. The work done by the friction, from the instant the block makes contact with the spring until it comes to a momentary halt, is -0.50 J.

hint:: kinetic energy = elastic potential energy + loss or $KE - \text{loss} = PE_{\text{el}}$

loss = 0.50 J (or you can say kinetic energy - 0.5 = elastic energy). Same thing.

solve for K. Then loss = friction $\times$ distance. distance is given. get friction.

friction = $\mu_k N$. $N = \text{weight}$. get μ_k .



(a) What is the spring constant of the spring?

(b) What is the coefficient of kinetic friction between the box and the rough surface?



1040N/m and 0.1



540N/m and 0.41



1040N/m and 0.20



1040N/m and 0.41



Question 16 6.25 pts

A spring-loaded dart gun is used to shoot a dart straight up into the air, and the dart reaches a maximum height of 24 meters above its point of release. The same dart is shot up a second time from the same gun, but this time the spring is compressed only half as far (compared to the first shot). How far up does the dart go this time? (Neglect friction and assume the spring is ideal and massless.)

hint: elastic potential energy = gravitational energy = mgh

Its the same spring and the same mass. so any mass, spring constant work. So take $K=1$ for example and mass = 1. if $h=24$ m. What's x ? Then What happens when x is divided by 2.



6.0 m



48 m



3.0 m



12 m

Not saved

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